Student ID _____ Student Name _____

IS222: Database Management Systems

Faculty of Science, Technology and Environment
School of Computing, Information and Mathematical Sciences

Final Examination
Semester I 2019

Face-To-Face and Online Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 180 minutes

INSTRUCTIONS

1. This examination has four sections:
   - Section A: Multiple Choice (20 marks)
   - Section B: Modelling & Normalization (40 marks)
   - Section C: SQL (25 marks)
   - Section D: Transaction (15 marks)
2. All questions are compulsory.
3. Write ALL answers in the space provided within this paper.
4. This exam is marked out of 100 and is worth 40% of your overall course marks.
5. To pass this paper you require at least 40% for this exam, which is equivalent to 40 marks out of 100.
6. This is a closed book exam.
7. This paper has 24 pages including this page.

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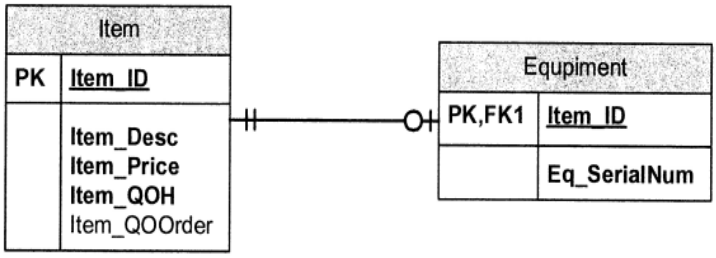
Section A: Multiple Choice 20 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question in the provided answer sheet on page 8. There is only **one** possible answer.

1) Which of the following is incorrect about business rules (BR)?
   a. A BR is bidirectional
   b. A BR states the cardinality of a relationship
   c. A BR states connectivity of a relationship
   d. A BR describes the relationship between two entities

2) Which of the following best describes the properties of the relationship shown below?



   a. Weak, Optional
   b. Strong, Optional
   c. Weak, Mandatory
   d. Strong, Mandatory

3) What does NoSQL mean?
   a. Not on SQL
   b. Not in SQL
   c. Not only SQL
   d. Note on SQL

4) Which of the following can be null?
   a. Primary Key
   b. Foreign Key
   c. Both (A) and (B) can be null
   d. Both (A) and (B) cannot be null

5) Which of the following is the same as a Row in a table?
    a. Attribute
    b. Tuple
    c. Relation
    d. Data

6) The _____ model is the end users' view of the data environment.
    a. Abstract
    b. External
    c. Conceptual
    d. Internal

7) Which key is solely responsible for data retrieval?
    a. Foreign Key
    b. Primary Key
    c. Super Key
    d. Secondary Key

8) The student table with a primary key STU_ID should not have which of the following dependency?
    a. Functional
    b. Partial
    c. Transitive
    d. All of the above

9) Given below is the ERD for the Patient table. Which attribute is a suitable candidate key

Patient	
PK	Pat_Code
	Pat_LName Pat_FName Pat_DOB Pat_Addr Pat_Phone

a. Pat_Lname
b. Pat_DOB
c. Pat_Addr
d. Pat_Phone

10) Which of these data types in MySql can be used to represent the value, AB23?

a. VCHAR
b. CHAR
c. STRCHAR
d. None of the above

11) To delete a row from a table, use the ____ command.

a. COMMIT
b. DELETE
c. DROP
d. ERASE

12) The ____ statement in SQL combines rows from two queries and returns only the rows that appear in the first set but not in the second.

a. UNION
b. UNION ALL
c. INTERSECT
d. MINUS

13) As long as two transactions, T1 and T2, access ____data, there is no conflict, and the order of execution is irrelevant to the outcome.

a. shared
b. common
c. unrelated
d. locked

14) One of the three most common data integrity and consistency problems is ____.

a. user errors
b. lost updates
c. disk failures
d. deadlocks

15) A consistent database is ____.

a. one in which all tables have foreign keys
b. one in which all tables are normalized
c. one in which all SQL statements only update one table at a time
d. one in which all data integrity constraints are satisfied

16) Granularity refers to ____.

a. the size of a table
b. the number of attributes in a table
c. the number of rows in a table
d. the level of detail represented by the values stored in a table's row

17) How many rows would be returned from a cross join of tables A and B, if A contains 8 rows and B contains 18?

a. 144
b. 26
c. 18
d. 8

18) ____ are required to prevent another transaction from reading inconsistent

a. Locks
b. Stamps
c. Logs
d. Schedules

19) A(n) ____ is an alternate name given to a column or table in any SQL statement.

a. data type
b. stored function
c. alias
d. trigger

20) Which of the following SQL statements will correctly create an index on the stud_lname field of the student table in a MySQL database.

a. `CREATE INDEX stud_lname ON student;`
b. `CREATE INDEX stud_idx_stud_lname ON stud_lname (student) ;`
c. `CREATE INDEX stud_idx studl_name FROM student (stud_lname) ;`
d. `CREATE INDEX stud_idx_stud_lname ON student (stud_lname);`

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Section A: Answer Sheet 20 Marks

Circle the correct answer.

1.	**a**	**b**	**c**	**d**
2.	**a**	**b**	**c**	**d**
3.	**a**	**b**	**c**	**d**
4.	**a**	**b**	**c**	**d**
5.	**a**	**b**	**c**	**d**
6.	**a**	**b**	**c**	**d**
7.	**a**	**b**	**c**	**d**
8.	**a**	**b**	**c**	**d**
9.	**a**	**b**	**c**	**d**
10.	**a**	**b**	**c**	**d**
11.	**a**	**b**	**c**	**d**
12.	**a**	**b**	**c**	**d**
13.	**a**	**b**	**c**	**d**
14.	**a**	**b**	**c**	**d**
15.	**a**	**b**	**c**	**d**
16.	**a**	**b**	**c**	**d**
17.	**a**	**b**	**c**	**d**
18.	**a**	**b**	**c**	**d**
19.	**a**	**b**	**c**	**d**
20.	**a**	**b**	**c**	**d**

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Section B: Modeling & Normalization 40 Marks

a) Give an example of a computerized file system and list any three of its major limitations? **(4 marks)**

b) What is structural dependence in the context of a file system? **(2 marks)**

c) List and explain the insert and update anomalies? **(3 marks)**

d) How do you decide whether a table has referential integrity? **(2 marks)**

e) Why are keys needed in a database? **(2 marks)**

f) List and explain any three features of a DBMS **(3 marks)**

g) How do you decide if an entity is weak or strong? Given an example of such an entity. **(3 marks)**

h) What is a derived attribute? Give relevant examples to support your answer **(3 marks)**

i) Discuss the difference between a composite key and a composite attribute. **(2 marks)**

j) Given the following business scenario, create the appropriate business rule(s). **(2 marks)**

> "Samsung has recently released their galaxy series 10 devices. Each Galaxy phone has its own specifications which includes details such as the storage capacity, processor speed, screen resolution and others. The phones are being sold at various retail stores across Fiji such as Vodafone, Digicel, Bondwell and others. In order to attract maximum customers, all retail stores sell a variety of phones."

k) Draw an Entity Relationship Diagram to depict the business rules mentioned above. You must include some attributes that you think should be part of each entity. **(4 marks)**

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l) What is the purpose of normalization? **(2 marks)**

m) Normalize the given table. You must show all the intermediary steps. **(8 marks)**

user_orders								
order_id	**sku**	**name**	**line_total**	**order_total**	**order_date**	**user_id**	**user_joined**	**user_address_state**
12345	S-12	Blue shirt	10.50	41.45	2015-01-01	123	2014-12-30	Delaware, New Jersey
12345	J-34	Windbreaker	23.00	41.45	2015-01-01	123	2014-12-30	Delaware, New Jersey
12345	B-13	Belt	7.95	41.45	2015-01-01	123	2014-12-30	Delaware, New Jersey
67891	J-35	Denim Jacket	21.00	36.75	2015-01-02	456	2014-12-31	Pennsylvania
67891	H-78	White Hat	15.75	36.75	2015-01-02	456	2014-12-31	Pennsylvania
111213	S-12	Blue shirt	10.50	28.50	2015-01-03	123	2014-12-30	Delaware, New Jersey
111213	H-78	White Hat	18.00	28.50	2015-01-03	123	2014-12-30	Delaware, New Jersey

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Section C: Structured Query Language (SQL) 25 Marks

Question 1

What are "referential constraint actions"? **(2 marks)**

Question 2

Explain why the following command would create an error, and what changes could be made to fix the error. **(2 marks)**

```
SELECT V_CODE, SUM(P_QOH) FROM PRODUCT;
```

Question 3

Suppose that you have two tables, EMPLOYEE and EMPLOYEE_1. The EMPLOYEE table contains the records for three employees: Alice Cordoza, John Cretchakov, and Anne McDonald. The EMPLOYEE_1 table contains the records for employees John Cretchakov and Mary Chen. Given that information, what is the query output for the UNION query? (List the query output.)

(2 marks)

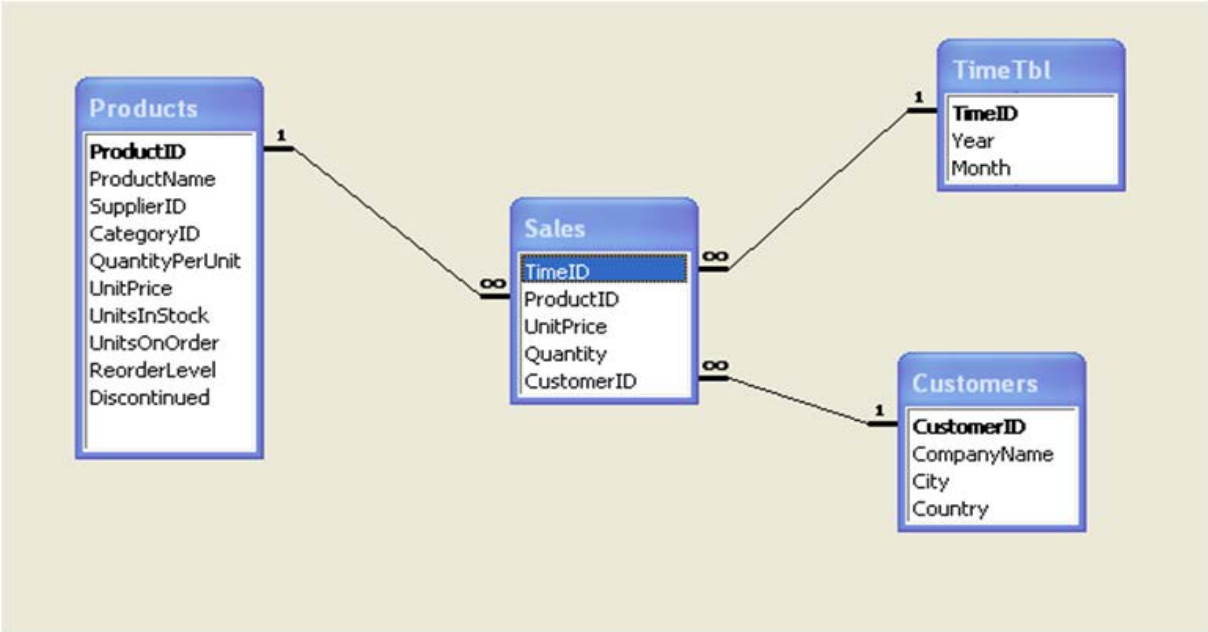
Question 4

What is query optimization? **(1 mark)**

Question 5

Assume the following database is used to store the sales data for a company. Use the following description and the database schema given on the next page to answer the following questions of this section.

The management team needs some reports for their upcoming meeting. In the following questions, you will be required to write the relevant queries that will provide the required information for their report. Your queries should be written for a MySQL database. You are required to use the given attribute names in the following ERD diagram in your queries.



a) A user named Sally needs to access the database. Write the necessary SQL command(s) so that Sally can login to her MySQL account given that her username is Sally and password is Secret. **(2 marks)**

b) Based on your experience gained from the IS222 lab exercises, Write the MySQL command that will create a database named Sales. **(1 mark)**

c) The structure of the Customers table is given in Figure 1 below. Write the necessary SQL Statement to create the Customers table **(4 Marks)**

Field	Type	Null	Key	Default
CustomerID	varchar(5)	NO	PRI	*NULL*
CompanyName	varchar(40)	NO	MUL	*NULL*
City	varchar(15)	YES	MUL	*NULL*
Country	varchar(15)	YES		*NULL*

Figure 1

d) List the country and the number of customers from that particular country **(1 mark)**

e) List the number and name of product whose category is 2, unit price is greater than 18 and product name not starting with "Chef". **(2 marks)**

f) List the number(s), name(s) and price of the most and least expensive product(s) **(3 marks)**

g) Update UnitsInStock to zero for whose product id is 33. **(2 marks)**

h) Add a column named “sum” whose data type is integer in products table. Update sum by adding the UnitsInStock and UnitsOnOrder whose supplier id and category id are the same. **(3 marks)**

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Section D: Transaction Processing 15 Marks

Question 1

What is transaction isolation? **(2 marks)**

Question 2

What is the purpose of a transaction log? **(2 marks)**

Question 3

List and discuss the five transaction properties. **(5 marks)**

Question 4

Consider the two transactions, T1 and T2, given below

```
T1:  Read1(A)
     Read1(B);
     If(A==0) Then B = B + 1
     Write1(B);

T2:  Read2(B);
     Read2(A);
     If (B==0) Then A = A + 1
     Write2(A);
```

Read1 and Read2 are read operations for their respective transactions, 1 and 2.

Write1 and Write2 are write operations for their respective transactions, 1 and 2.

Assume that both A & B are 0 initially with either A=0 or B=0. Answer the questions that follow in the spaces given:

a) Will serial execution of the transactions above always preserve consistency? Explain your answer?

(2 marks)

b) Identify all conflicting operations during concurrent execution of the transactions**.**

(1 mark)

Question 5

Consider that there are only two transactions, T1 and T2, with timestamps TS1 and TS2, respectively as shown in table 1 below. X, Y and Z are data items. Assume that T2 first requests for exclusive lock on Y, at the same time, T1 requests for exclusive lock on X. Then T1 next requests a lock on Y.

Transaction T2 TS2 "2019-04-15 22:50:56"	**Transaction T1 TS1** "2019-04-15 22:51:12"
	Read (X)
	Read (Y)
Read (Y)	
Read (Z)	
	Write (X)
Write (Y)	
Write (Z)	

Table 1

Explain specifically how the locks will be acquired/released by the transactions from the outset and how the transactions will come to completion using the Wait-die algorithm. **(3 marks)**

THE END

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